DSE-BYTOPIC
MATH EP
M2

Please stick the barcode label here.

Candidate Number							

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION BYTOPIC

MATHEMATICS Extended Part
Module 2 (Algebra and Calculus)
Question-Answer Book

Time allowed : $2\frac{1}{2}$ hours

This paper must be answered in English

INSTRUCTIONS

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1.
(2) This paper consists of TWO sections, A and B.
(3) This paper carried 100 marks.
(4) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
(5) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
(6) Unless otherwise specified, all working must be clearly shown.
(7) Unless otherwise specified, numerical answers must be exact.
(8) The diagrams in this paper are not necessarily drawn to scale.
(9) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

FORMULAS FOR REFERENCE

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$	$\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$	$\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	$\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$
$2\sin A \cos B = \sin(A+B) + \sin(A-B)$	$\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$
$2\cos A \cos B = \cos(A+B) + \cos(A-B)$	
$2\sin A \sin B = \cos(A-B) - \cos(A+B)$	

****

SECTION A (63 marks)

1. (a) Prove by mathematical induction that $\displaystyle\sum_{k=1}^{n} \frac{k(k-1)}{2} = \frac{(n-1)n(n+1)}{6}$ for all positive integers n. (4 marks)

   (b) Evaluate $\displaystyle\sum_{k=1}^{10} \frac{k(k-1)}{2}$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. (a) Prove by mathematical induction that $\sum_{i=1}^{n} \frac{1}{i^2+i} = \frac{n}{n+1}$ for all positive integers n.

(4 marks)

(b) Evaluate $\sum_{i=1}^{99} \frac{1}{i^2+i}$.

(2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. Let $\{a_n\}$ and $\{b_n\}$ be two sequences of positive integers. It is given that for all positive integers n,

$$a_n = \sum_{k=1}^{n} (2^k + 1)$$

and $\{b_n\}$ satisfies the recurrence relation $b_1 = 3$ and $b_{n+1} = \frac{1}{2} b_n + a_n$ for $n \geq 1$.

(a) Prove by mathematical induction that for all positive integers n,

$$a_n = 2^{n+1} + n - 2.$$

(4 marks)

(b) Prove by mathematical induction that for all positive integers n,

$$b_n = \frac{1}{3} \left(2^{n+2} + 6n - 24 + \frac{19}{2^{n-1}} \right).$$

(4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. (a) Prove by mathematical induction that $\displaystyle\sum_{r=1}^{n} \frac{r}{r^4+r^2+1} = \frac{n(n+1)}{2(n^2+n+1)}$ for all positive integers n. (4 marks)

(b) Evaluate $\displaystyle\sum_{r=33}^{64} \frac{r}{r^4+r^2+1}$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. (a) Prove by mathematical induction that $\sum_{i=1}^{n}\left(\frac{1}{2}\right)^{-i}\cdot i^3 = 2^{n+1}(n^3-3n^2+9n-13)+26$ for all positive integers n. (4 marks)

(b) Evaluate $\sum_{i=8}^{13}\left(\frac{1}{2}\right)^{-i}\cdot i^3$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. (a) Prove by mathematical induction that $\sum_{i=1}^{n} \frac{3i^4 + i^6}{(-1)^i} = \frac{(-1)^n}{2} n^3 (n+1)^3$ for all positive integers n. (4 marks)

(b) Evaluate $\sum_{i=9}^{16} \frac{3i^4 + i^6}{(-1)^{i+1}}$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. (a) Prove by mathematical induction that $\displaystyle\sum_{n=1}^{k} \frac{1}{n(n+1)(n+2)} = \frac{k(k+3)}{4(k+1)(k+2)}$ for all positive integers k. (4 marks)

(b) Using (a), prove that $\displaystyle\sum_{n=1}^{k} \frac{2n+k}{n(n+1)(n+2)} = \frac{k+4}{4} - \frac{5k+4}{2(k+1)(k+2)}$ for all positive integers k. (4 marks)

(c) Evaluate $\displaystyle\sum_{n=6}^{18} \frac{2n+21}{(n+1)(n+2)(n+3)}$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. (a) Prove by mathematical induction that $\sum_{k=1}^{n}\left(\frac{1+\sqrt{5}}{2}\right)^{k-1}\left(k\left(\frac{1+\sqrt{5}}{2}\right)-k+1\right)=n\left(\frac{1+\sqrt{5}}{2}\right)^{n}$ for all positive integers n. (4 marks)

(b) Evaluate $\sum_{k=1}^{10^{10}}\left(\frac{1+\sqrt{5}}{2}\right)^{(10^{10})}-\left(\frac{1+\sqrt{5}}{2}\right)^{k-3}\left(\left(\frac{1+\sqrt{5}}{2}\right)+1\right)\left(k\left(\frac{1+\sqrt{5}}{2}\right)-k+1\right)$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. Let $\{u_n\}$ be a sequence whose first few terms are defined by $u_1 = 1$, $u_2 = 5$, and for all positive integers n, satisfies the following recurrence relation:

$$u_{n+2} = 5u_{n+1} - 6u_n$$

(a) Find the values of u_3 and u_4. (2 marks)

(b) Prove by mathematical induction that for all positive integers n,

$$u_n = 3^n - 2^n.$$

(4 marks)

(c) Let $S_n = \sum_{r=1}^{n} u_r$.

(i) By considering the result of (b), prove that $S_n = \dfrac{3^{n+1} - 2^{n+2} + 1}{2}$. (1 mark)

(ii) Prove by mathematical induction that for all positive integers n, $3^{n+1} - 2^{n+2} + 1$ is divisible by 2. (4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

SECTION B (37 marks)

10. (a) Prove by mathematical induction that $\displaystyle\sum_{k=1}^{n} \frac{1}{(k+1)\sqrt{k}+k\sqrt{k+1}} = 1 - \frac{1}{\sqrt{n+1}}$ for all positive integers n. (4 marks)

(b) Evaluate $\displaystyle\sum_{k=1225}^{2024} \frac{1}{(k+1)\sqrt{k}+k\sqrt{k+1}}$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. (a) Prove by mathematical induction that $\sum_{k=1}^{n} \tan k\theta \tan(k-1)\theta = \frac{\tan n\theta}{\tan\theta} - n$ for all positive integers n. (4 marks)

(b) Solve $\sum_{k=1}^{6} \tan(k+6)\theta \tan(k+5)\theta = -6$, where $0 < \theta < \frac{\pi}{2}$. (3 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. (a) Prove by mathematical induction that for all positive integers n,

$$\sum_{k=1}^{n} \frac{\sin 2k\theta}{\cos(2k-1)\theta\cos(2k+1)\theta} = \frac{\sec(2n+1)\theta - \sec\theta}{2\sin\theta}.$$

(4 marks)

(b) Evaluate $\displaystyle\sum_{k=1}^{256} \frac{\sin\frac{k\pi}{8}}{\cos\frac{(2k-1)\pi}{16}\cos\frac{(2k+1)\pi}{16}}$. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. Let $B = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}$ and $A = 4I + B$, where I is the 2×2 identity matrix.

(a) Find B^2. (1 mark)

(b) Prove by mathematical induction that $A^n = 4^n I + n4^{n-1}B$ for all positive integers n. (4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

14. Let $A = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 0 & 2 & 2 \end{pmatrix}$.

Prove by mathematical induction that $A^n = \begin{pmatrix} 2^n & 0 & 0 \\ n2^n & 2^n & 0 \\ n(n-1)2^{n-1} & n2^n & 2^n \end{pmatrix}$ for all positive integers n.

(4 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. In a plane, a sequence of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \ldots, \mathbf{v}_n$ is defined by

$$\mathbf{v}_n = n\mathbf{i} + (n+1)\mathbf{j}$$

where $\mathbf{i}$ and $\mathbf{j}$ are unit vectors along the x-axis and y-axis respectively.

A scalar P_n is defined as the sum of dot products of the first n vectors and their subsequent vectors:

$$P_n = \sum_{r=1}^{n} (\mathbf{v}_r \cdot \mathbf{v}_{r+1}).$$

(a) (i) Calculate the value of the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$. (1 mark)

(ii) Hence, find the values of P_1 and P_2. (2 marks)

(b) Prove by mathematical induction that for all positive integers n,

$$P_n = \frac{n(n+1)(2n+7)}{3}.$$

(4 marks)

END OF PAPER

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.